

Computer Science & IT

Database Management System



Transactions & concurrency control

Lecture No. 04



By- Vishal Sir



Recap of Previous Lecture



✓ Topic

Equivalent schedule

✓ Topic

Serializable schedule

✓ Topic

RW problem

✓ Topic

WR problem

Topics to be Covered



✓
Topic

WW problem

✓
Topic

Lost update problem

Topic

Classification of schedules

Topic

Classification based on recoverability



Topic : Problems because of concurrent execution

① R-W Problem / Read-Write Problem / Write after Read Problem

② W-R Problem / Write-Read Problem / Read after Write problem

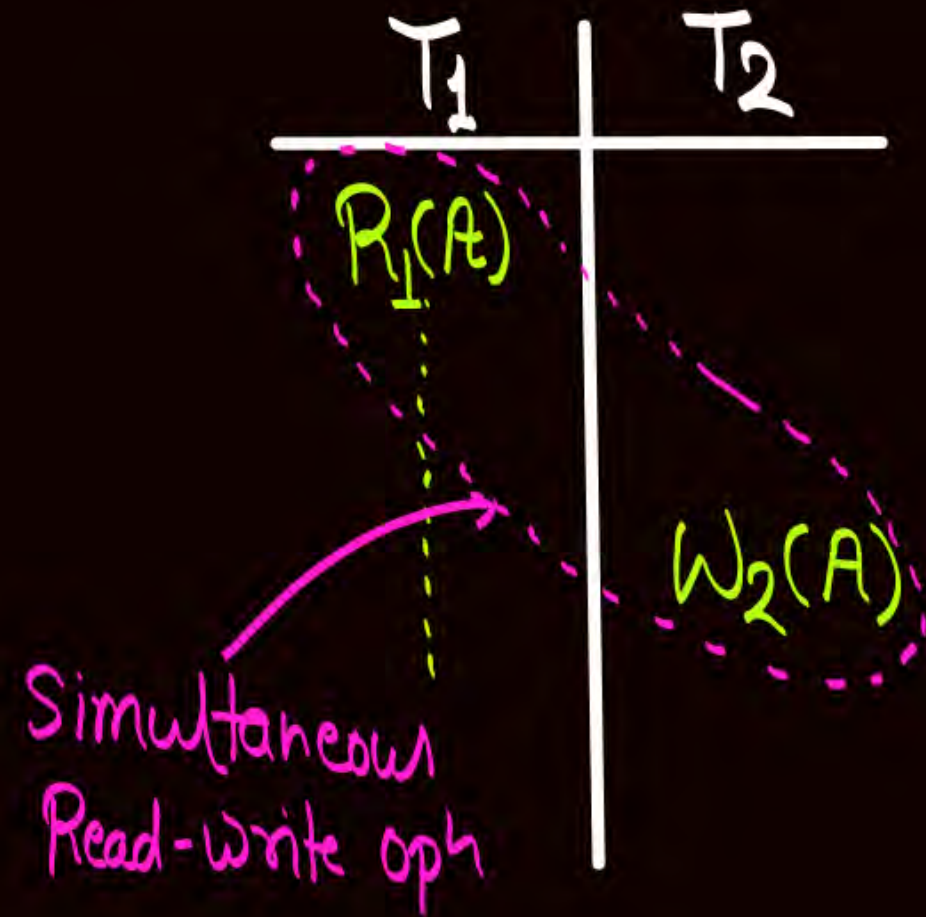
③ W-W Problem / Write-Write Problem / Write after Write Problem

④ Lost update problem

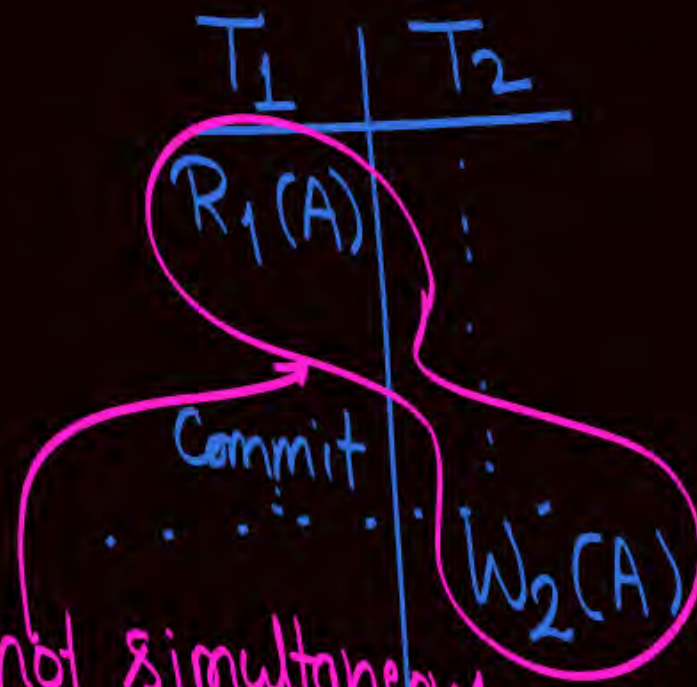
This problems exists only if the schedule is a non-serializable schedule.

It is possible with both serializable as well as non-serializable schedules

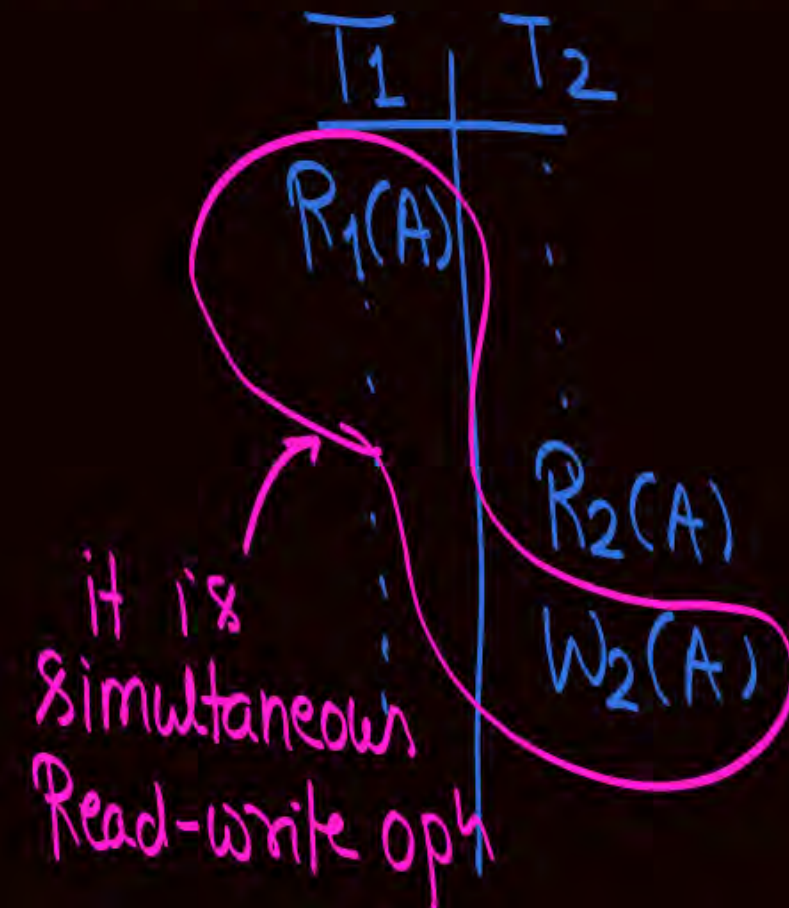
RW Problem / Read-Write Problem / Write-after-Read Problem



→ If transaction T_2 updates the data item 'A' which is already read by an uncommitted transaction T_1 , then such sequence of opⁿ is called **Simultaneous Read-write opⁿ**



it is not simultaneous Read-write opⁿ because T_1 has already Committed

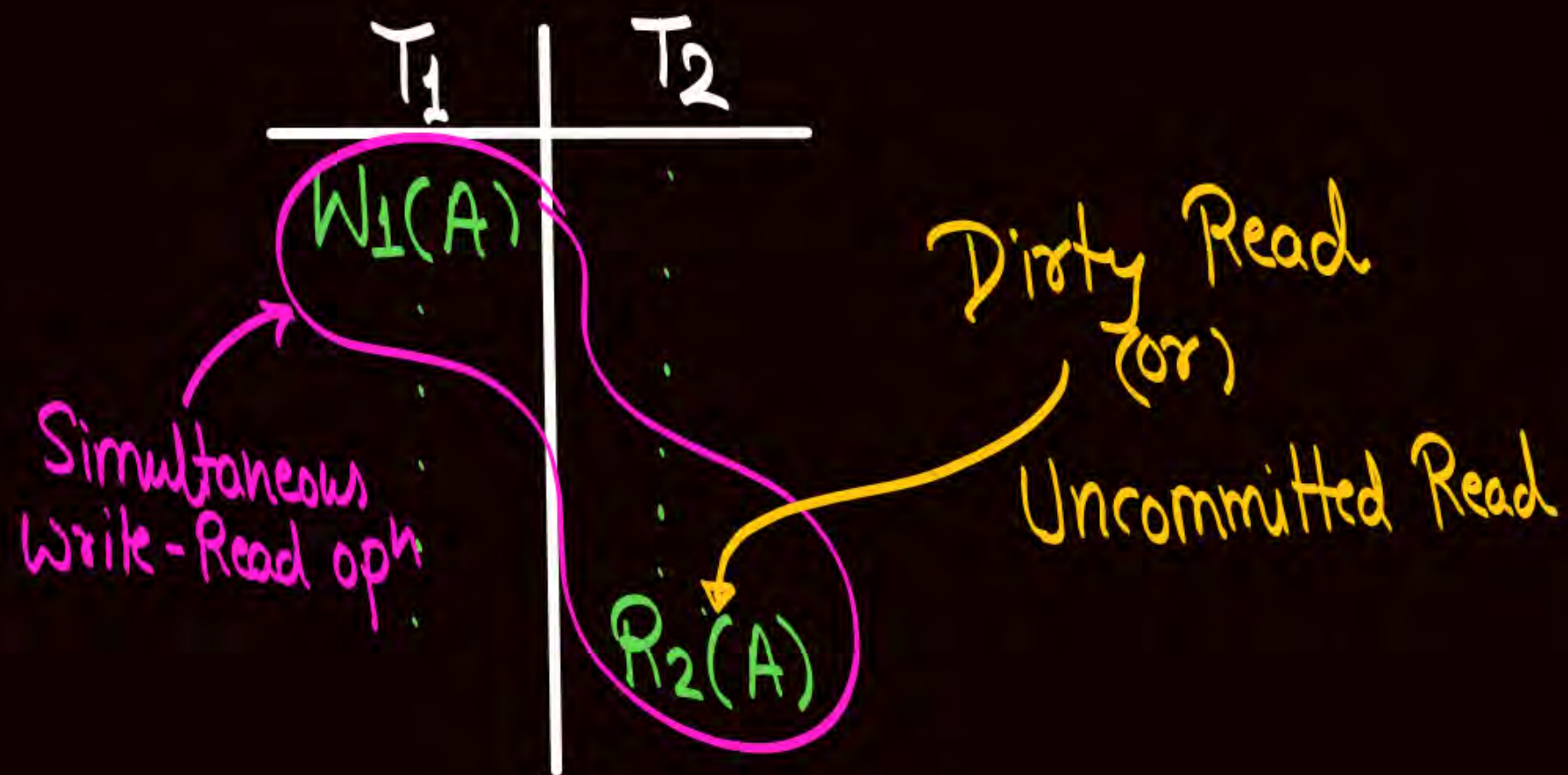


RW Problem / Read-Write Problem / Write-after-Read Problem

→ A schedule is said to have Read-Write problem if and only if

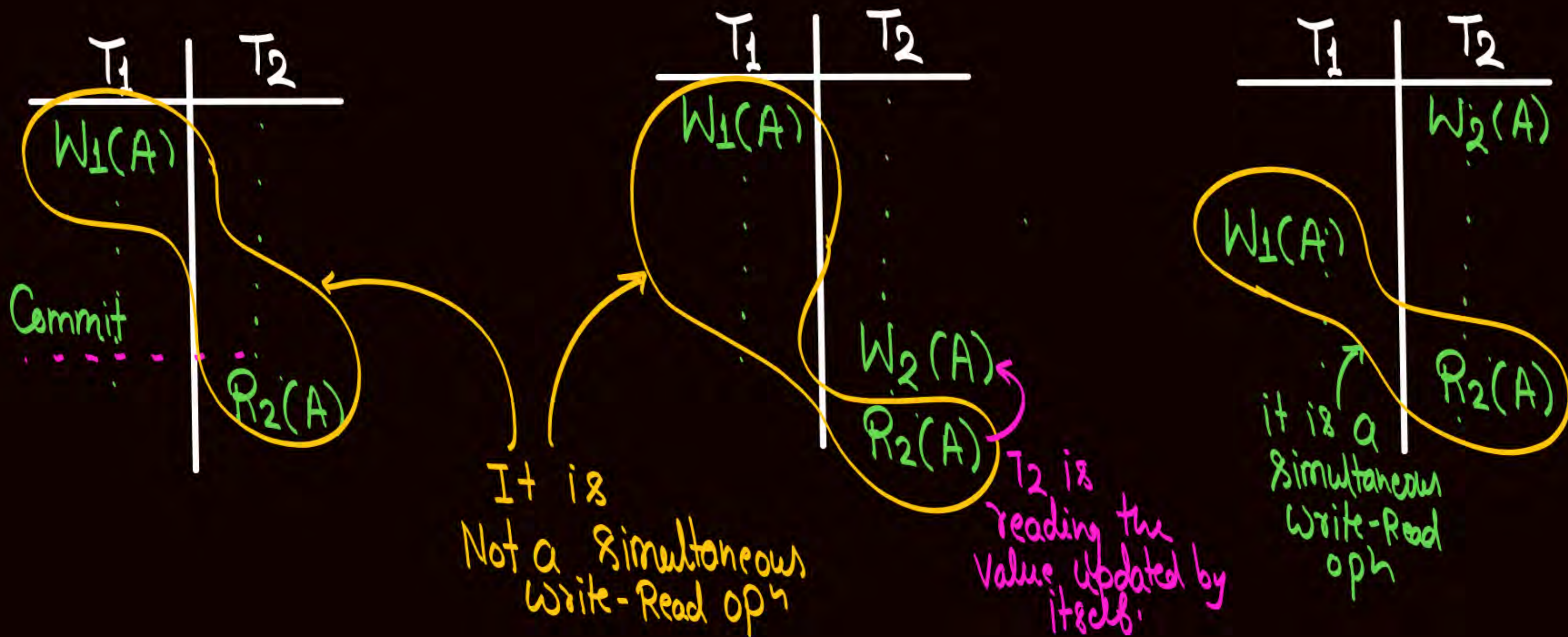
- and
- ① Simultaneous Read-write opⁿ must exist
 - ② Schedule must be a non-serializable schedule

WR Problem / Write-Read Problem / Read-after-write Problem
also known as "Dirty Read Problem"



If transaction T_2 reads the value updated by an uncommitted transaction T_1 , then that sequence of opⁿ is called a Simultaneous Write-Read opⁿ. {And, that read opⁿ is called} an uncommitted read

WR Problem / Write-Read Problem / Read-after-write Problem
also known as "Dirty Read Problem"



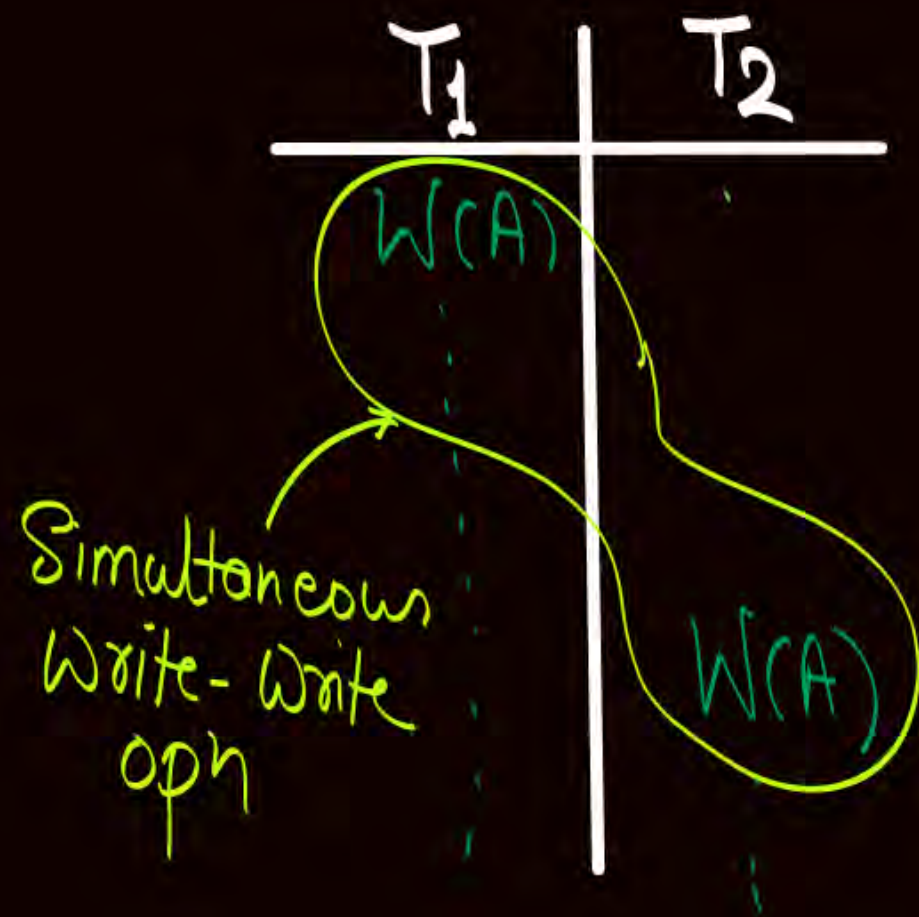
WR Problem / Write-Read Problem / Read-after-write Problem
also known as "Dirty Read Problem"

* A schedule is said to have Write-Read problem if and only if

① Simultaneous Write-Read opⁿ must exist.
{i.e.; Uncommitted Read opⁿ must exist}

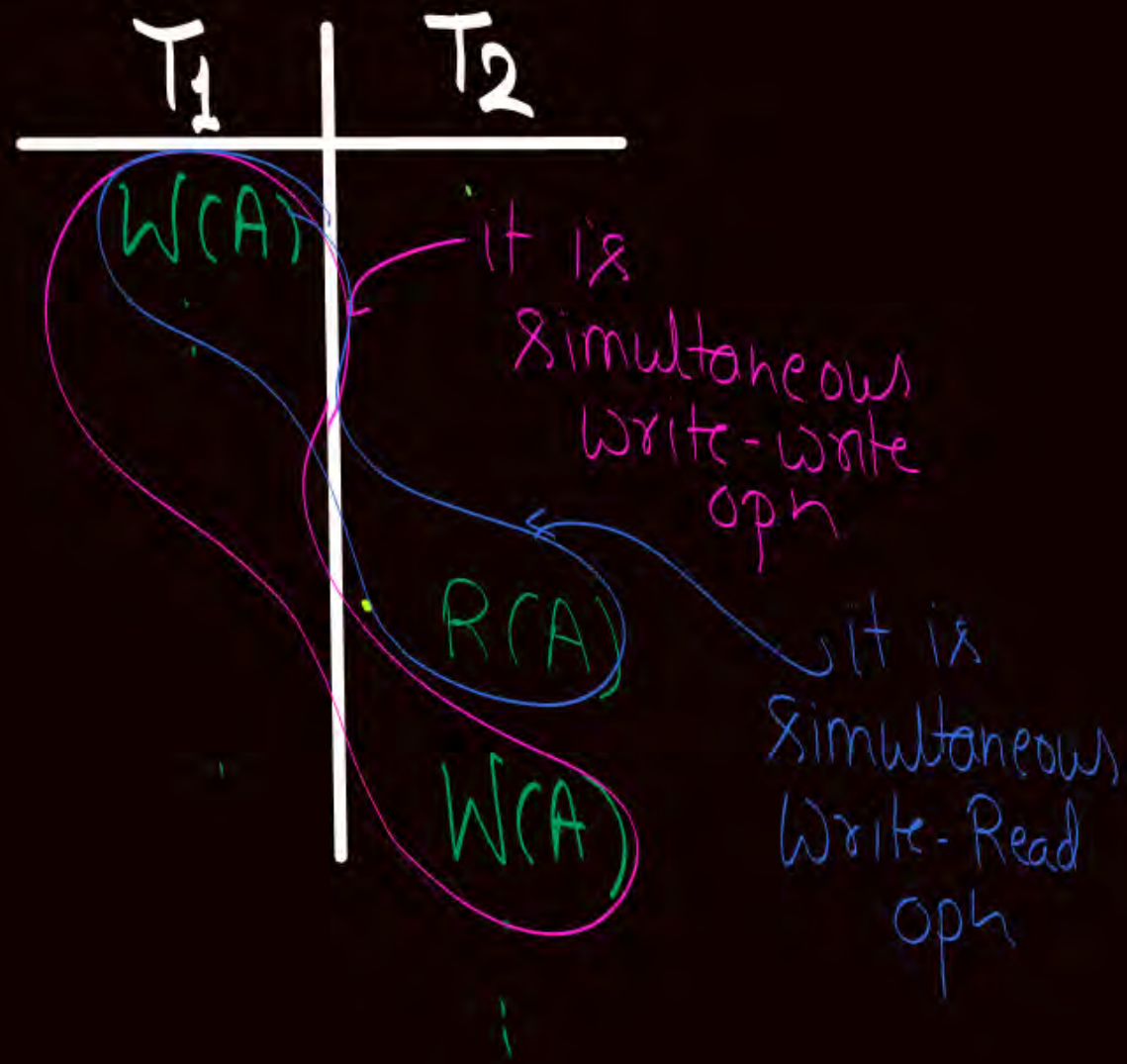
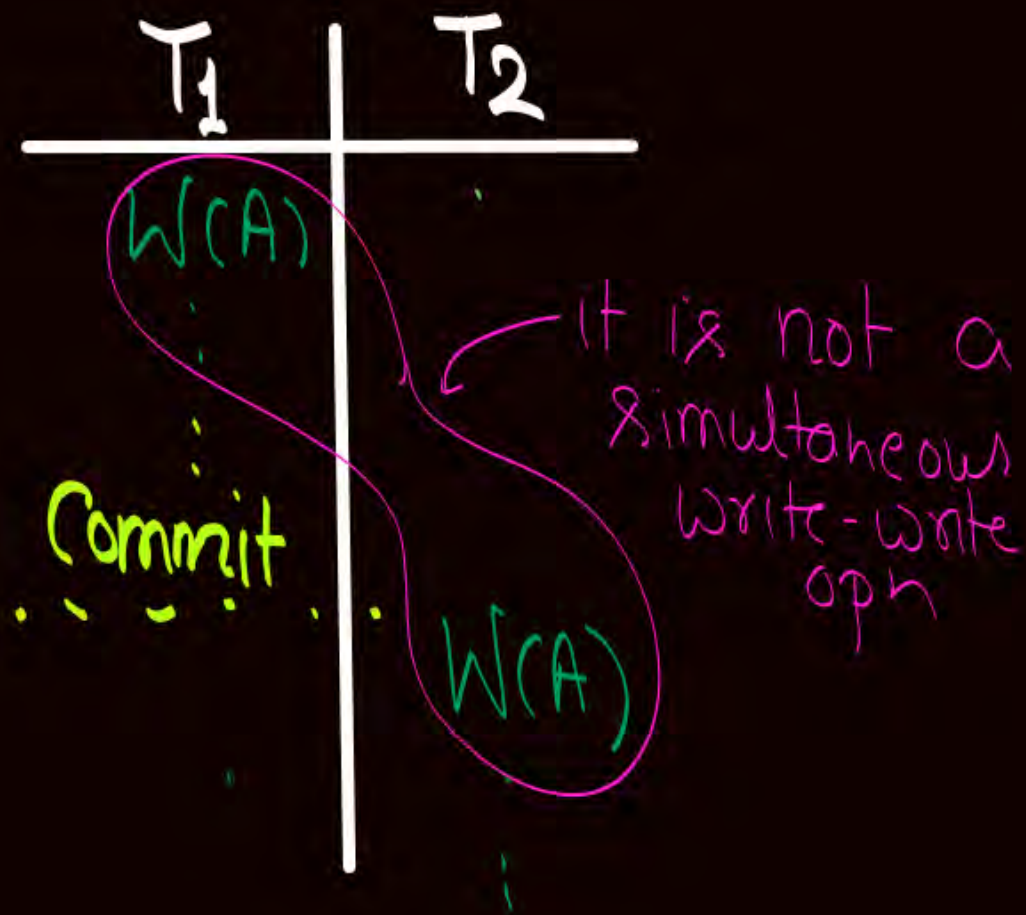
And ② Schedule must be a non-serializable schedule

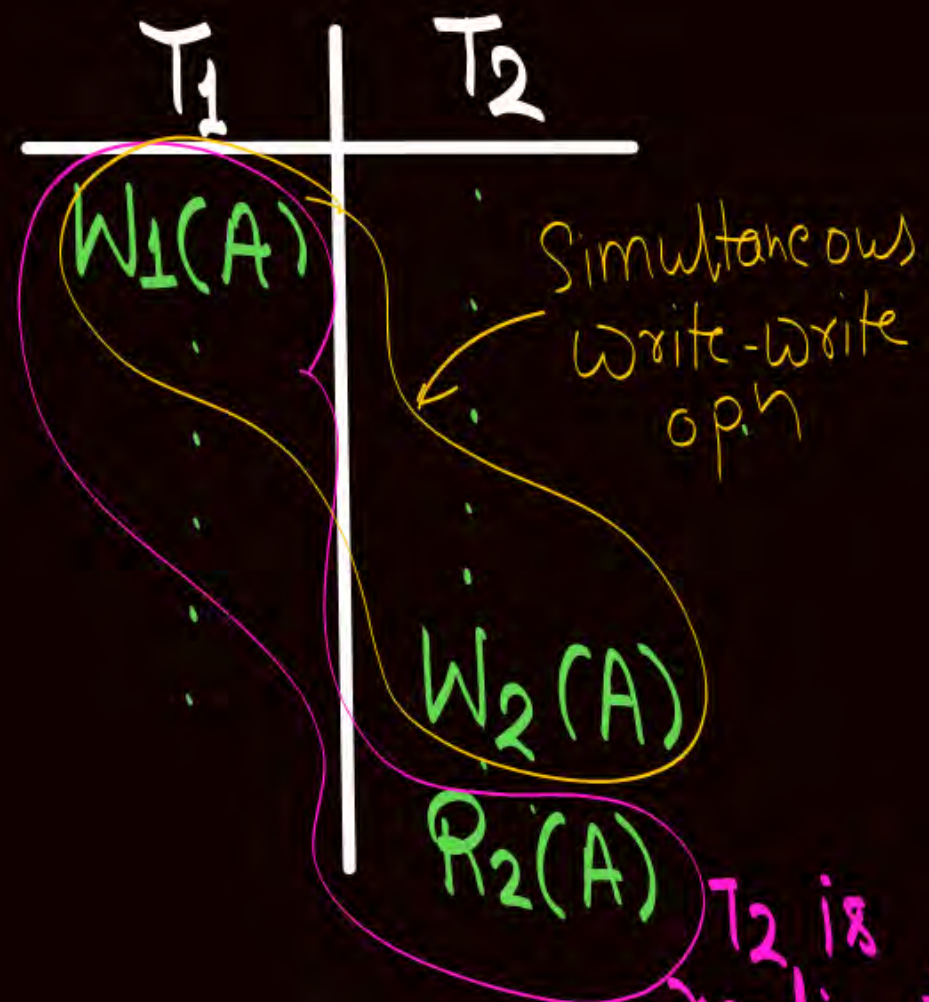
WW Problem / Write-Write Problem / Write-after-write Problem



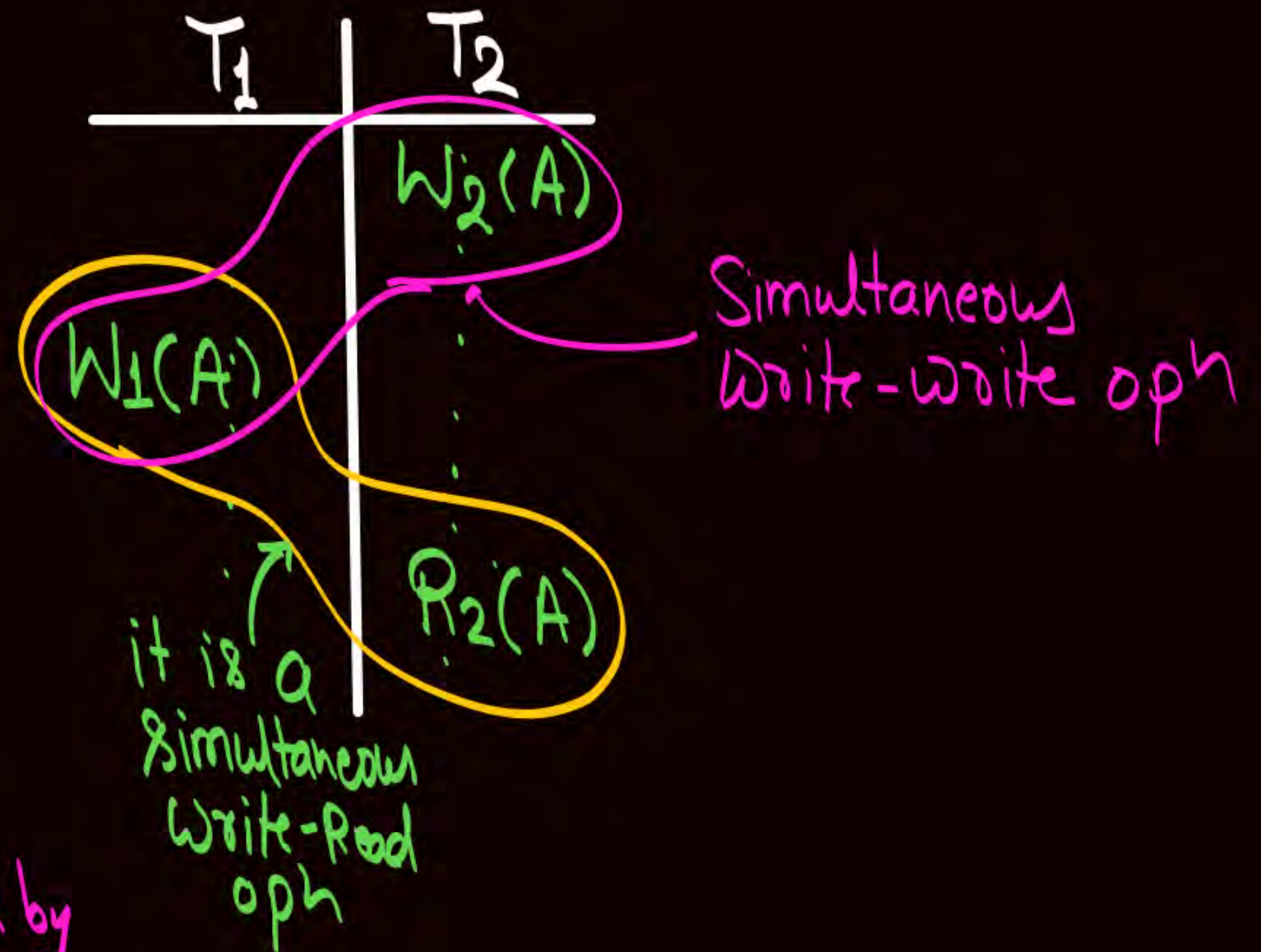
If transaction T_2 updates the data item 'A' which is already updated by an uncommitted transaction T_1 , then that sequence of opⁿ is called simultaneous write-write opⁿ.

WW Problem / Write-Write Problem / Write-after-write Problem





T₂ is reading the value updated by itself.
 ∴ Not a Simultaneous Write-Read opⁿ



WW Problem / Write-Write Problem / Write-after-write Problem

A schedule is said to have write-write problem if and only if

- ① Simultaneous write-write opⁿ must exist
- and ② Schedule must be a non-serializable schedule

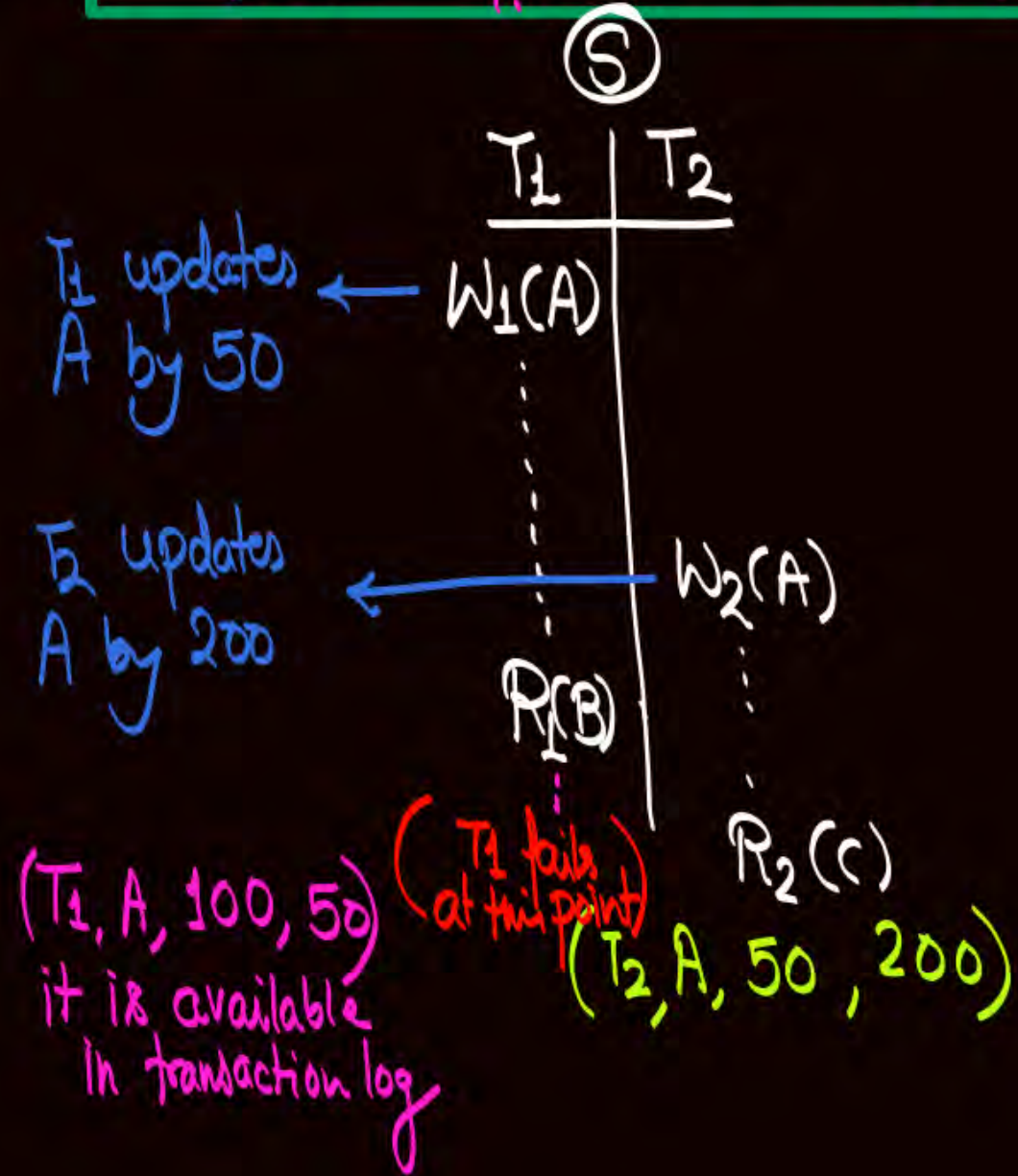
Note ∴ If schedule is a serializable schedule
then none of RW Problem or WR problem
or WW problem can exist,

but lost update problem is still possible

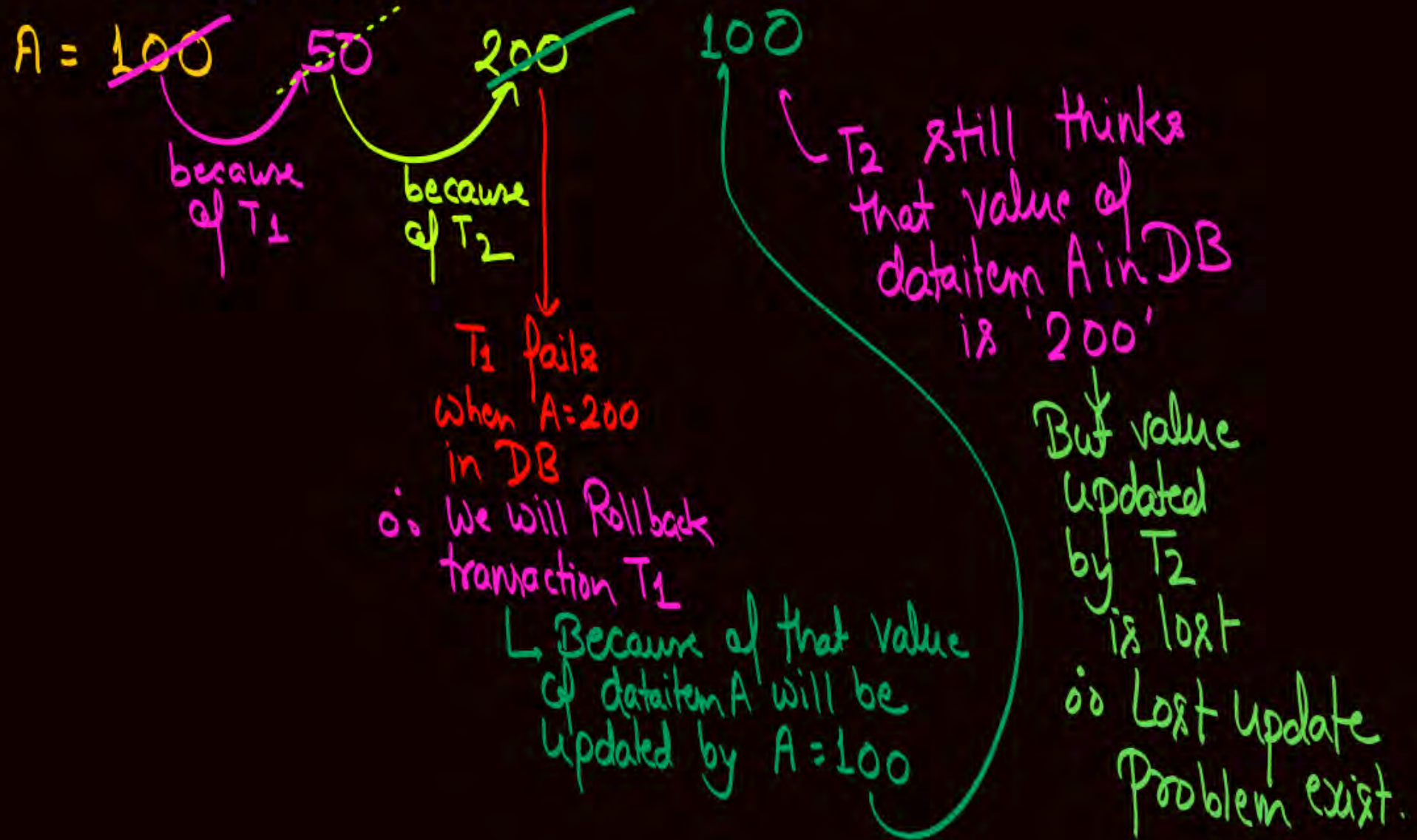
Lost update Problem

A schedule is said to have lost update problem if and only if simultaneous write-write opn exists in the schedule, does not matter whether the schedule is a serializable schedule or not.

* Lost update Problem



Let initially the value of dataitem A in the DB = 100

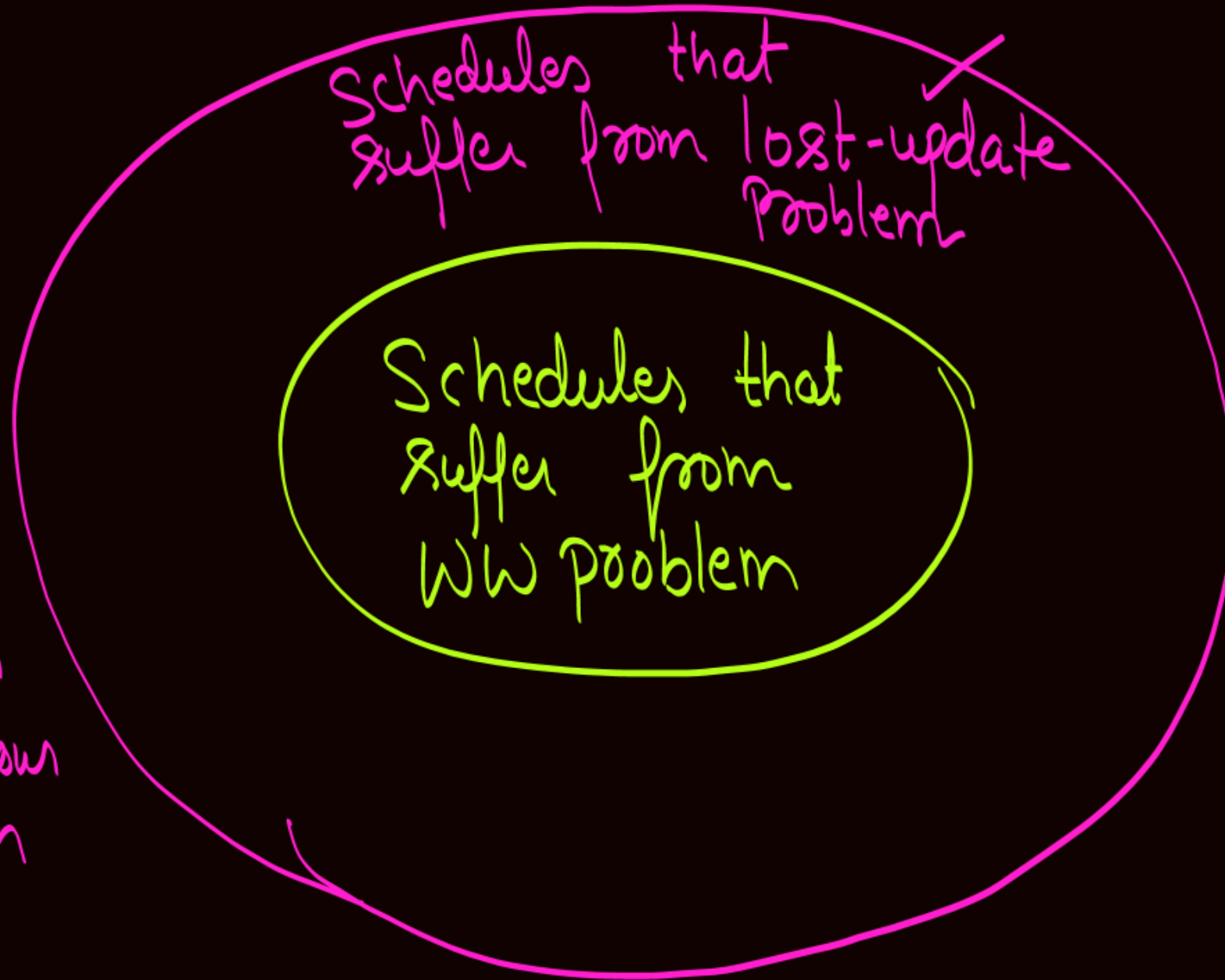


Note:-

Schedules that
suffer from lost-update
problem ✓

Schedules that
suffer from
WW problem

There are
all the schedules
with simultaneous
write-write op



H.W.

Q:- Check whether the schedule is a serializable schedule or not?

④

S ₄	
T ₁	T ₂
R ₁ (A)	
	R ₂ (B)
W ₁ (B)	
	W ₂ (B)

$$S_4 \neq T_1 \rightarrow T_2$$

$$S_4 \neq T_2 \rightarrow T_1$$

∴ S₄ is a non-serializable schedule



Topic : Classification of schedule

Classification based on Recoverability

- ① Irrecoverable Schedule
- ② Recoverable Schedule
- ③ Cascadeless Rollback Recoverable Schedule
- ④ Strict Recoverable Schedule

Classification based on Serializability

- ① Conflict Serializable Schedule
- ② View Serializable Schedule

Classification based on
Recoverability

Note :-

If transaction T_j reads the value written by an uncommitted transaction T_i , then transaction T_j is dependent on transaction T_i .

If transaction T_i rollback for some reason then we will have to rollback all the transactions that are dependent on transaction T_i { for our example it is T_j }

T_i	T_j
$W_i(A)$	
	$R_j(A)$

T_j reads the value written by an uncommitted transaction T_i
{ T_j is dependent on T_i }

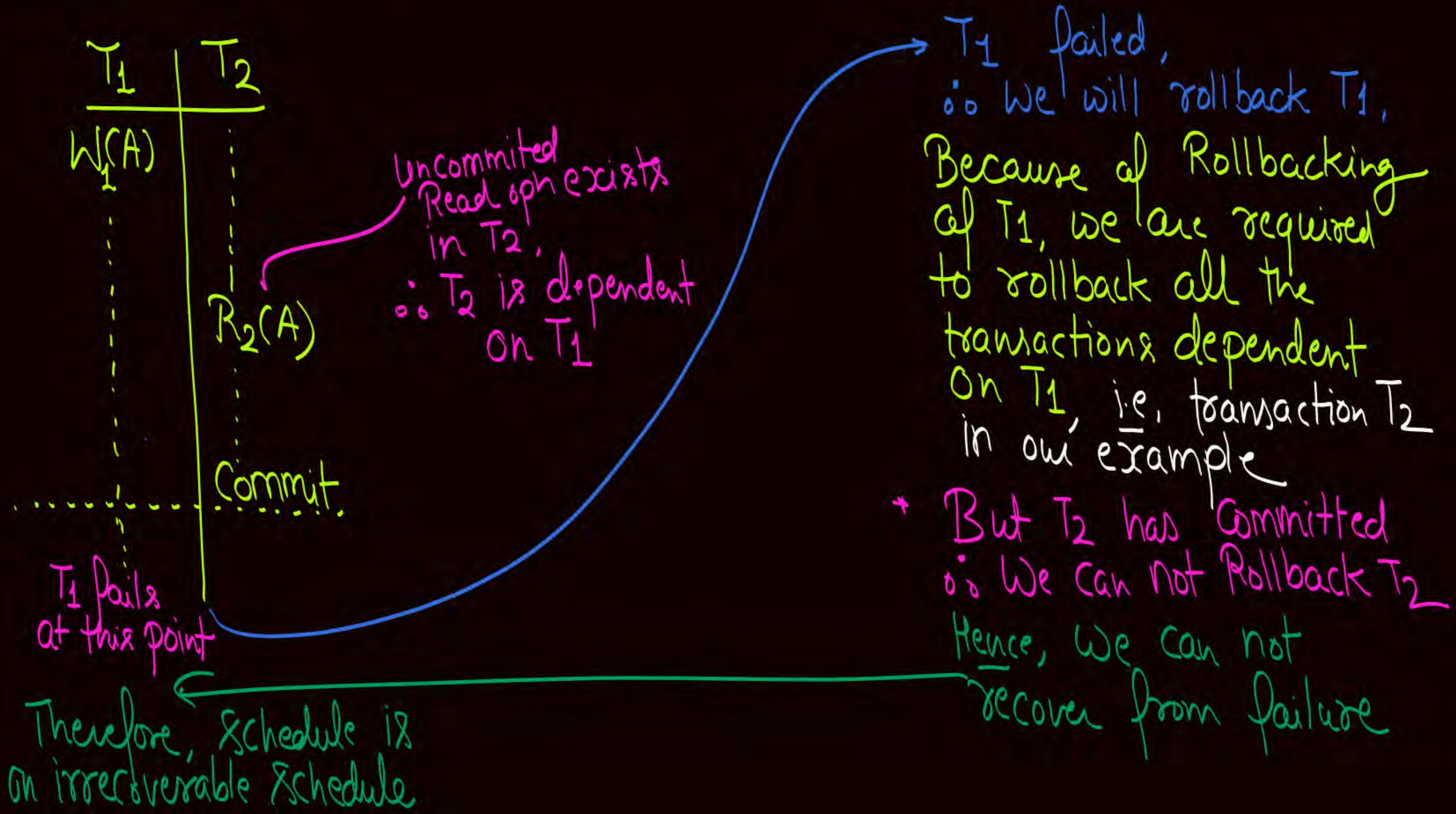
We can rollback a transaction only if that transaction has not yet committed, Rollbacking of a committed transaction is not allowed



Topic : Irrecoverable schedule

- * It is not possible to rollback a Committed transaction.
- * If there arise a situation when we are required to rollback a Committed transaction, then it will not be possible to rollback that transaction, and hence we will not be able to recover from that failure. Such Schedules are called irrecoverable schedule.

Irrecoverable Schedule



Irrecoverable Schedule



uncommitted
opⁿ

$R_2(A)$

T_2 is dependent
on T_1

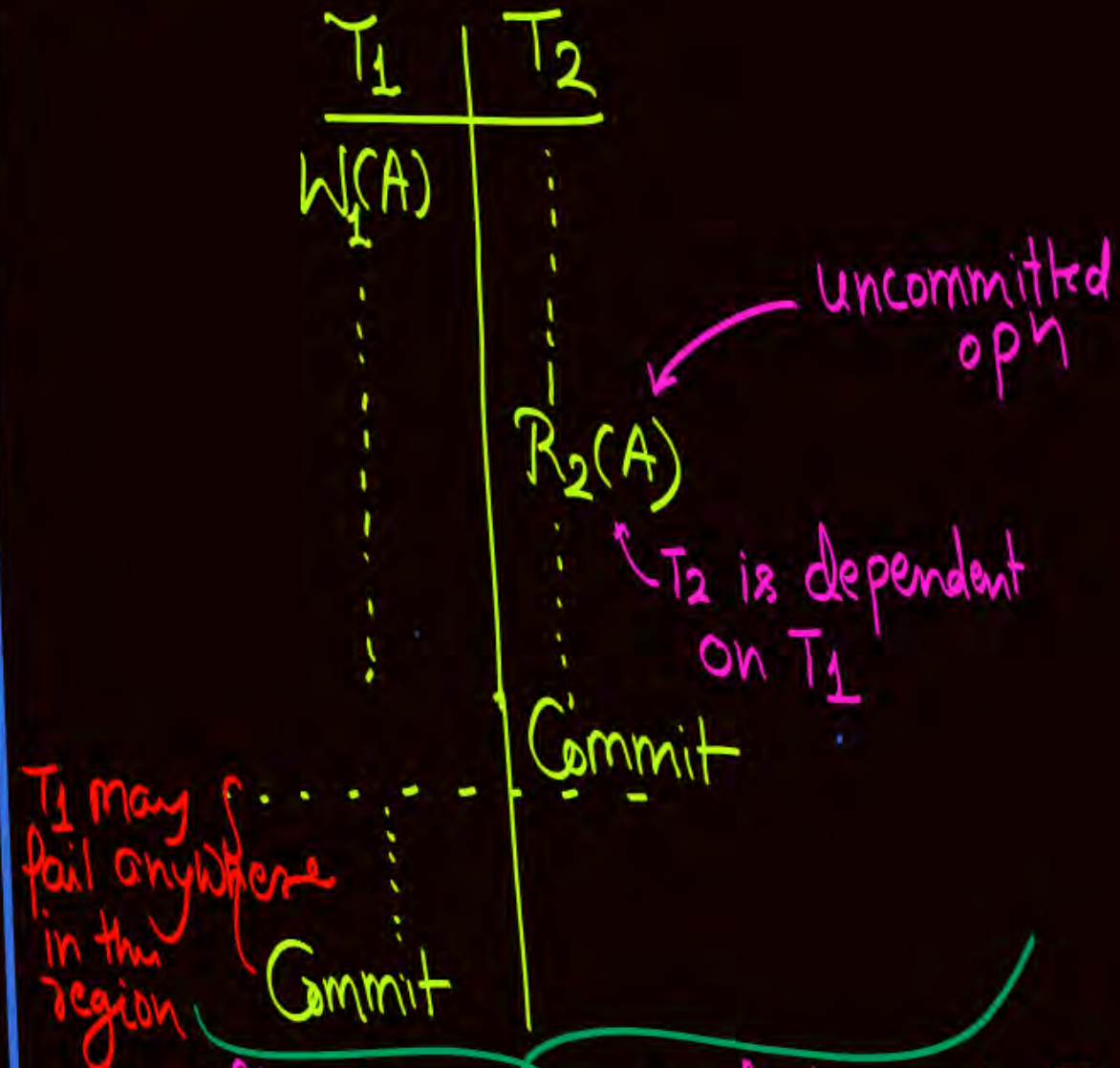
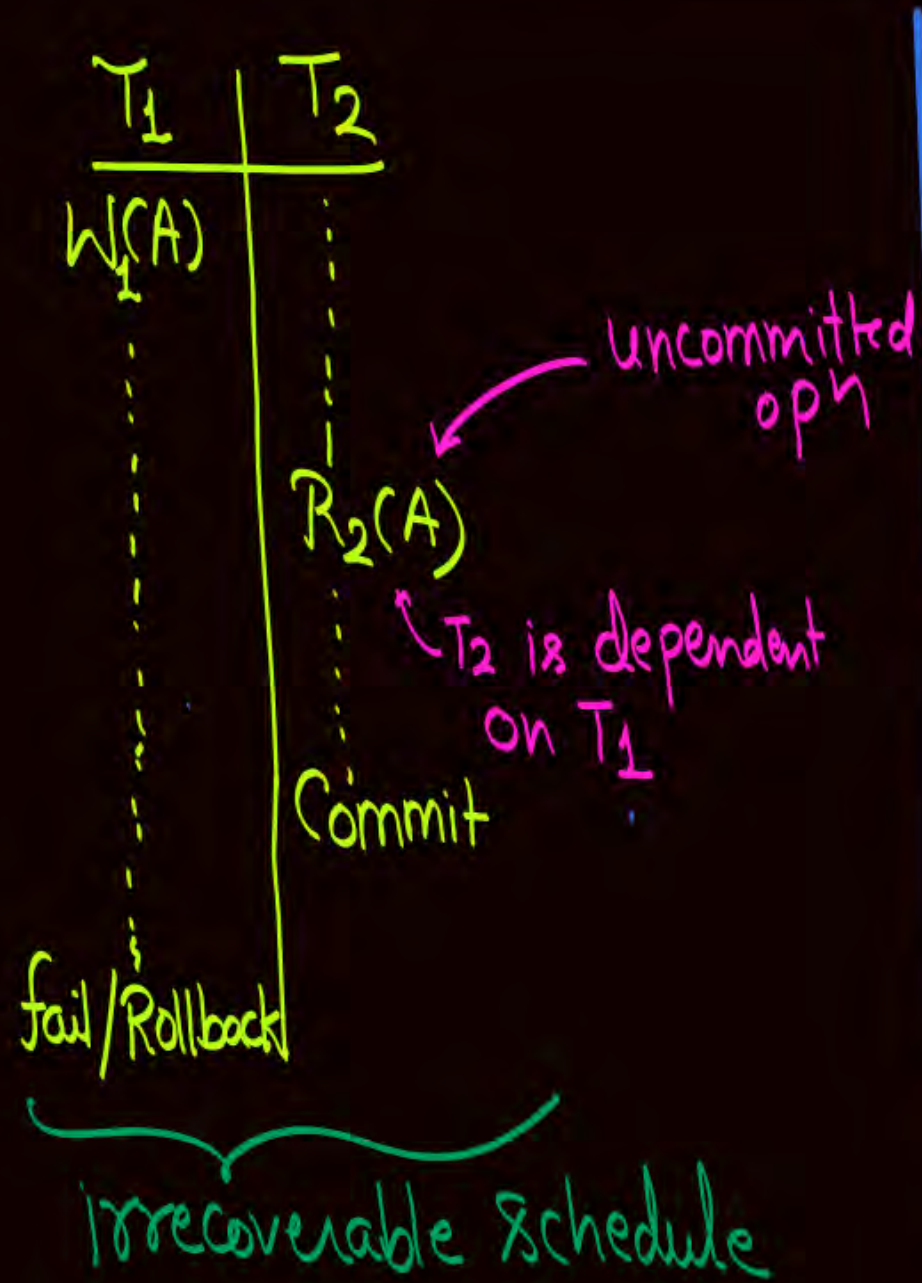
No transaction has performed
the commit opⁿ.

∴ If failure occur then
both can rollback

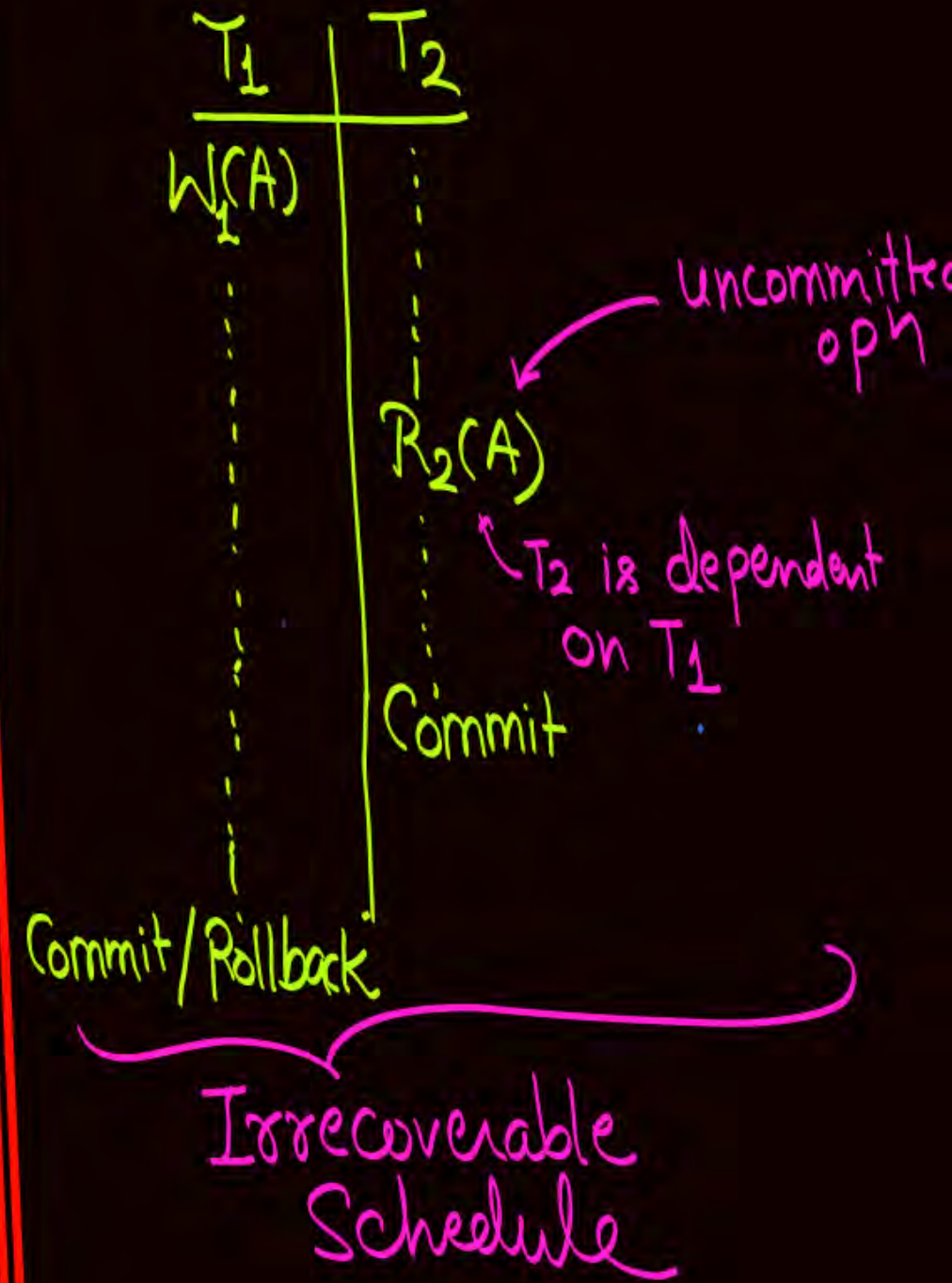
ie, we will be able to recover
from failure

∴ Schedule is a
Recoverable Schedule

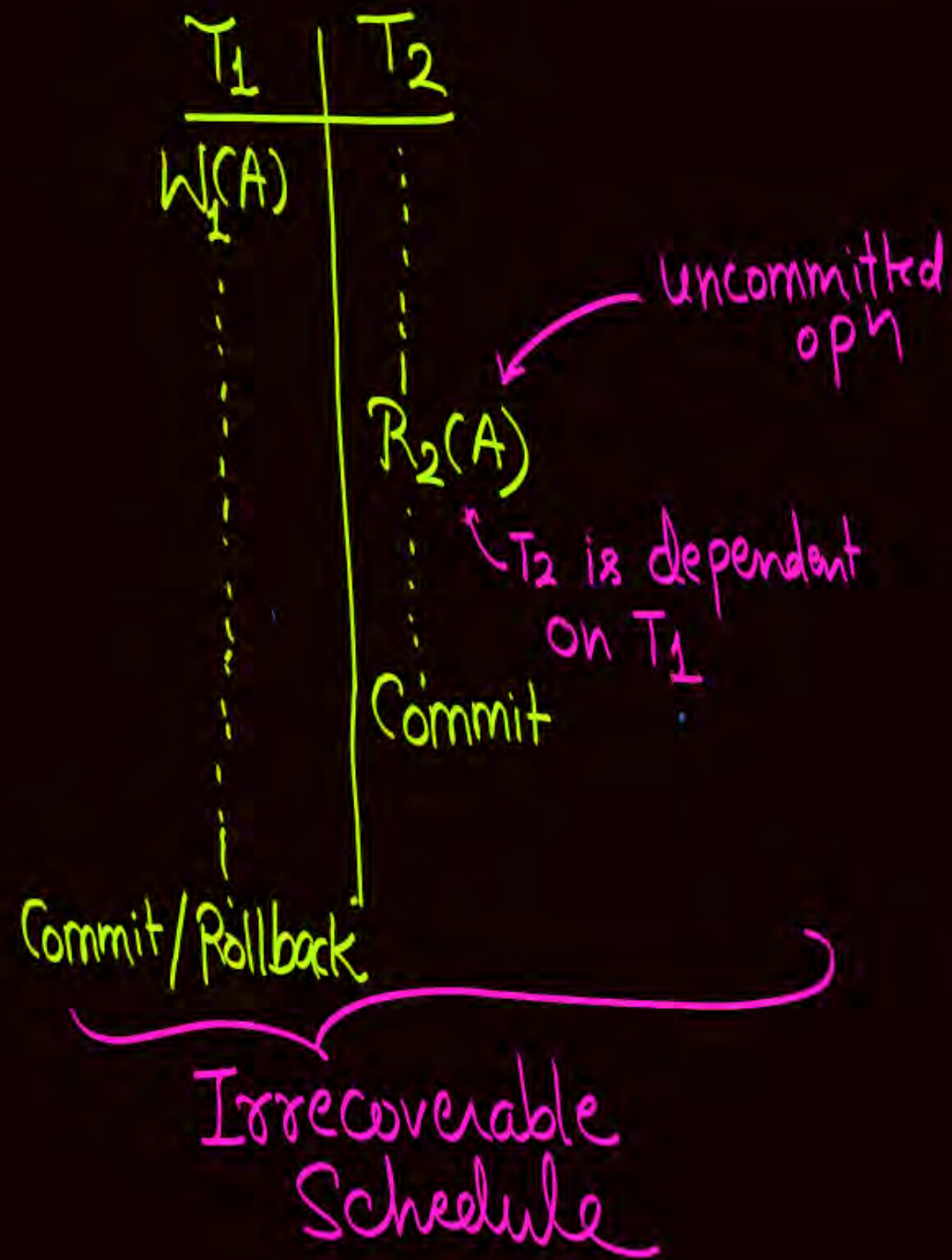
Irrecoverable Schedule



After the Commit of transaction T₂, schedule is unsafe because T₁ may fail, and if it happens we will not be able to recover
∴ it is also irrecoverable schedule



Irrecoverable Schedule



Let two transaction T_1 and T_2 exists in a schedule and

- If transaction T_2 is dependent on transaction T_1 , and if transaction T_2 (i.e. dependent transaction) Commit before Commit or Rollback of transaction T_1 (i.e. the transaction on which it depends), then that schedule is called irrecoverable schedule.

Irrecoverable Schedule

→ Uncommitted Read opⁿ are necessary (not sufficient) for a schedule to be irrecoverable

• i.e., ① If there is no uncommitted read opⁿ in the schedule then schedule can not be irrecoverable {i.e., it is recoverable}

Ans ② If uncommitted opⁿ exists in the schedule, then schedule may or may not be irrecoverable.



Topic : Recoverable schedule

→ For a schedule to be called a recoverable schedule either uncommitted Read opⁿ should not exist in the schedule

(or) if uncommitted read opⁿ exist in the schedule, then dependent transaction must commit after the Commit/Rollback of the transactions on which they depend



Recoverable Schedule



Topic : Cascading rollback problem

→ If failure of one transaction results in rolling back of a set of other transactions along with rolling back of failed transaction, then we say that cascading rollback problem exists in the schedule



Topic : Cascading rollback problem

Cascading Rollback Problem results in wastage of CPU time & IO time



T ₁	T ₂	T ₃	T ₄	T ₅
W(A)	R(A)	R(A) W(A)	R(A)	R(A)

* T₁ fails

No one has committed
∴ All can rollback

Hence Recoverable Schedule
let T₁ fails at this point

- T₁ fails, ∴ We will rollback T₁
- Because of Rollbacking of T₁ we need to rollback all the transactions dependent on T₁, i.e., T₂ & T₃
- Because of rollbacking of T₂ & T₃ We need to rollback the transactions dependent on T₂ & T₃, No transaction is dependent on T₂, but T₄ & T₅ are dependent on T₃, ∴ We need to rollback T₄ & T₅ as well.

Because of Rollback of one transaction we had to rollback multiple transaction,
∴ Cascading Rollback Problem exist



Topic : Cascadeless recoverable schedule

→ If uncommitted Read opⁿ exists in the schedule, then dependency among the transactions will exist, and if dependency exists among the transactions, then because of failure of one transaction we will have to rollback all dependent transactions and hence cascading rollback problem will exist.

∴ For Cascadeless recoverable schedule, uncommitted operations must not be allowed.



2 mins Summary



✓ **Topic**

WW problem

✓ **Topic**

Lost update problem

✓ **Topic**

Classification of schedules

Topic

Classification based on recoverability



THANK - YOU